

Business Problem Statement

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A leading logistics and transportation company wants to improve the efficiency, reliability, and cost-effectiveness of its vehicle fleet. Management needs better visibility into vehicle utilization, maintenance costs, fuel consumption, downtime, trip performance, and driver performance.

The company is particularly interested in identifying vehicles with high maintenance costs or downtime, understanding fuel consumption patterns, monitoring fleet utilization, and evaluating driver and operational performance.

Overarching Business Question

“How can the company use fleet, maintenance, fuel, trip, and driver data to reduce operational costs, minimize vehicle downtime, improve fleet utilization, and enhance overall fleet performance?”

Deliverables

1. Data Preparation & Modeling (Python) Clean, transform, validate, and prepare the raw fleet datasets for analysis using Python and Pandas.
2. Data Analysis (SQL) Organize the data into a structured MySQL database and perform SQL analysis using queries, CTEs, window functions, and views to identify fleet, maintenance, fuel, and driver performance insights.
3. Visualization & Insights (Power BI) Build interactive dashboards showing executive KPIs, fleet performance, maintenance trends, fuel costs, vehicle utilization, downtime, and driver performance.
4. Report and Recommendations Summarize the major findings and provide practical recommendations for improving fleet efficiency, controlling maintenance and fuel costs, and reducing downtime.
5. GitHub Repository Maintain a well-structured repository containing the Python ETL scripts, SQL queries, Power BI dashboard, documentation, and project screenshots.